# DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026

X UNIT TEST

Subject: CHEMISTRY

## SECTION A [5]

1.Which one of the following precipitates are formed when carbon dioxide gas is passed through lime water?

(a) $Ca(OH)_2$ (b) $Ca(OH)_4$ (c) $CaCO_3$ (d) $CaCO_2$

2. What is the colour of the ash formed when a magnesium ribbon is burnt in the air?

(a) White (b) Black (c) Yellow (d) Pink

3. Heating of ferrous sulphate is a type of :

(a) Decomposition reaction (b) Combination Reaction

(c) Displacement reaction (d) All of the above

**Following questions consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below:**

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4.**Assertion:** Photosynthesis is considered as an exothermic reaction.

**Reason:** Photosynthesis is an endothermic reaction because sunlight energy is absorbed by green plants during this process.

5.**Assertion:** Chemical equations should be balanced.

**Reason:** As per the law of conservation of mass, mass can neither be created nor be destroyed.

## SECTION B [6]

6. A white salt on heating decomposes to give brown fumes and a residue is left behind.

(a) Name the salt.

(b) Write the equation for the decomposition reaction.

7.List four observations that help us to determine whether a chemical reaction has taken place.

8.When a solution of potassium iodide is added to a solution of lead nitrate in a test tube, a reaction takes place.

(a) What type of reaction is this?

(b) Write a balanced chemical equation to represent the above reaction.

## SECTION C

[4]

**9. Read the passage carefully and answer the following questions from (i) to (iv).**

The substance which we use for white-washing our houses is calcium oxide. We put calcium oxide in adrum and add water to it slowly. Calcium oxide reacts with water vigorously to form a white solid calledslaked lime with the evolution of heat. More water is then added to get slaked lime solution.(This slakedlime solution is then applied to the walls of the house with the brush. Since, this process gives a white,shiny appearance to the walls of a house, it is called white-washing and gives a shiny finish to the walls)

(i)What is the chemical formula of slaked lime?

(ii) Name the chemical reaction resulting to form slaked lime.

(iii)Give the chemical equation for the formation of slaked lime.

(iv) What is the reason behind shining appearance of walls after white washing?

## SECTION D

[5]

**10. Translate the following statements into chemical equations and balance them.**

(i) Hydrogen sulphide reacts with sulphur dioxide to form sulphur and water.

(ii) Methane on burning combines with oxygen to produce carbon dioxide and water.

(iii) Aluminium reacts with copper(II) sulphate to form aluminium sulphate and copper.

(iv) Calcium carbonate on heating gives calcium oxide and carbon dioxide.

(v)Barium chloride reacts with sodium sulphate to form barium sulphate and sodium chloride.

# DELHIPUBLICSCHOOLJANKIPURAM,LUCKNOW

## SESSION2025-2026:UNIT TEST

**Subject:Science : Physics(086)**

**TIME :45 min** **M.M. : 20**

**Date :**_____**Name :** _____**Class/SecX**_____

**General Instructions:**

**Readthefollowinginstructionscarefullyandfollowthem:**

- **Q1toQ5 carryonemark each.**
- **Q6toQ8areveryshorttypequestionsandcarry two markseach.**
- **Q9arecasebasedquestionandisoffourmarks each.**
- **Q10 arelonganswertypequestionsandcarryfivemarks each.**

**********

1. When a plane mirroris rotated through a certain angle, the reflected ray turns through twice asmuch and the size of the image:
   a) is doubled
   (b) is halved
   (c) becomes infinite
   (d) remains the same

2. An object is kept at a distance more than twice the focal length (F) from a concave mirror. The distance the image formed will be
   (a) less then F
   (b) equal to F
   (c) between F and 2 F
   (d) More than 2 F

3. No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be
   (a) Plane
   (b) Concave
   (c) Convex
   (d) Either concave or convex.

   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true but R is not the correct explanation of A.
   (c) A is true but R is false.
   (d) A is false but R is true.

4. **Assertion(A) :** The centre of curvature is not a part of the mirror. It lies outside its reflecting surface.
   **Reason (R) :** The reflecting surface of a spherical mirror forms a part of a sphere. This sphere has a centre.

b

5. **Assertion(A) :** For observing traffic at back, the driver mirror is convex mirror.
**Reason (R) :** A convex mirror has much larger field of view than a plane mirror.

6. Draw a labeled ray diagram to show the path of the reflected ray corresponding to an incident ray of light parallel to the principal axis of a convex mirror. Mark the angle of incidence and angle of reflection on it.

7. Name the type of mirrors used in the design of solar furnaces. Explain how high temperature is achieved by this device.

8. State the two laws of reflection of light.

9. The linear magnification produced by a spherical mirror is +1/3. Analyzing this value state the (i) type of mirror and (ii) position of the object with respect to the pole of the mirror. Draw any diagram to justify your answer.

10.(a)Suppose you have three concave mirrors A, B and C of focal lengths 10 cm, 15 cm and 20 cm. For each concave mirror you perform the experiment of image formation for three values of object distances of 10 cm, 20 cm and 30 cm. By giving reason, answer the following:
(a) For the three object distances, identify the mirror/mirrors which will form an image ofmagnification -1.
(b) Out of the three mirrors, identify the mirror which would be preferred to be used for shaving purposes/make up.
(c) For the mirror B draw ray diagram for image formation for object distances 10 cm and 20 cm.

(b) Define the following terms in the context of spherical mirrors:
(i) Pole
(ii) Centre of curvature
(iii) Principal axis
(iv) Principal focus



250 copies

# DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

## SESSION 2025-2026 : UNIT TEST

**SET–I**

**Subject : Science (Biology)**

**TIME : 45 mins.** **M.M. : 20**

**Date :**_____ **Name :** _____ **Class/Sec X**_____

**General Instructions:**

**Read the following instructions carefully and follow them:**

- Q1 to Q5 are MCQs, carry one mark each.
- Q6 to 8 are short answer type questions and carry two marks each.
- Q9 is diagram based and carry four marks.
- Q10 is long answer type question and carry five marks.

**********

## Section A

1. Rings of cartilage present in the throat ensure that
(a) the air passage collapses (b) the air passage does not collapse
(c) the air is filtered entering the nostrils (d) the food passage does not collapse

**2. Which one of the following juices secreted in the human body does not contain any enzyme?**
(a) salivary juice (b) bile juice (c) gastric juice (d) pancreatic juice

3.Where does the light-dependent reaction of photosynthesis take place?
(a) Stroma (b) Cytoplasm (c) Grana (d) Mitochondria

For each question 4 and 5, select the correct answer from the following options:
(A) Both Assertion and Reason are correct, and Reason explains Assertion.
(B) Both Assertion and Reason are correct, but Reason does not explain Assertion.
(C) Assertion is correct, but Reason is incorrect.
(D) Both Assertion and Reason are incorrect.

4. Assertion: The rate of photosynthesis increases with increasing light intensity.
Reason: At low light intensity, photosynthetic activity is limited.

5. Assertion: The walls of the intestine have numerous villi.
Reason: These villi increase the surface area for absorption of nutrients.

## Section B

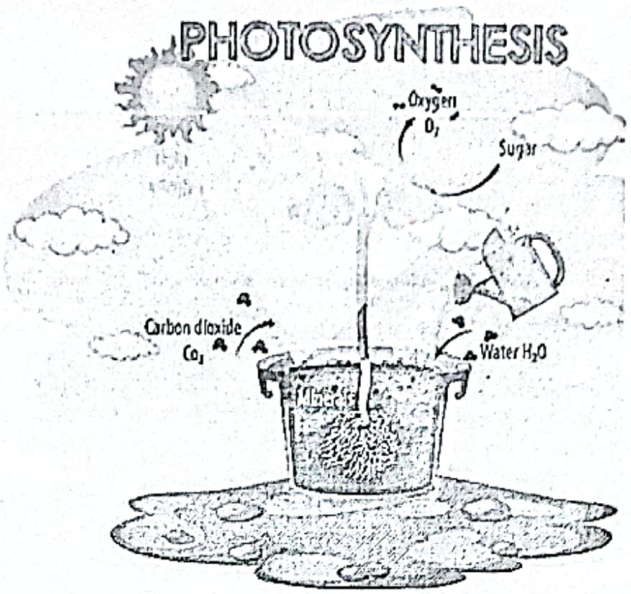
6. Draw a flow chart showing breakdown of glucose in aerobic and anaerobic condition.

7. We often hear people complain about 'acidity' in the stomach.
(a) Overproduction of what substance is most likely the reason for the complaint?
(b) Why is the production of this substance necessary?

8. Where does gaseous exchange take place in plants? Draw the well labelled diagram of that part.

## Section C

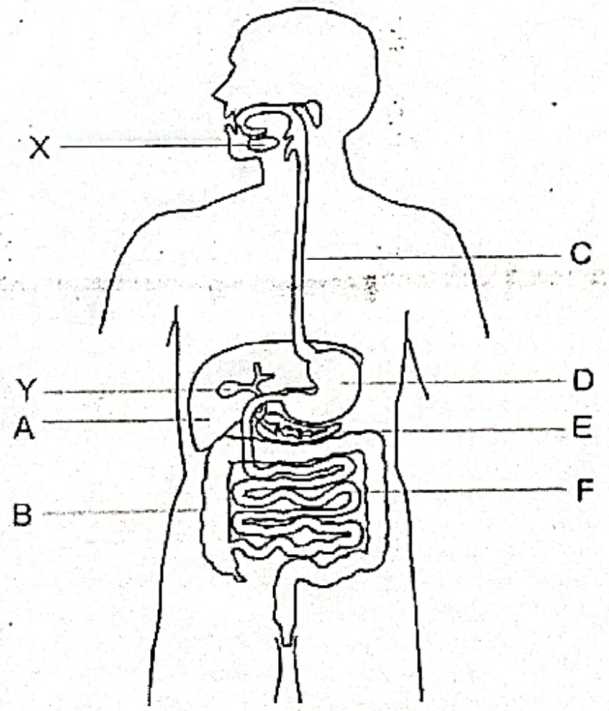
9. The following diagram represents the process of photosynthesis. Observe the same and answer the following questions. 1+1+2




a. The organelle where photosynthesis occurs. chloroplast
b. write the chemical equation of photosynthesis.
c. The raw materials required for photosynthesis.
$CO_2$, water, sun, chlorophyll

## Section D

10. This diagram shows the human digestive system. 3+1+1




(a) Name the parts A to F.
(b) Name the part where protein digestion starts.
(c) Name the part which shows peristalsis.